

Comparative Analysis of CNN From Scratch and MobileNetV2 Transfer Learning for Banana Ripeness Classification

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Submitted : **Abstrak–Background:** Accurate assessment of banana ripeness is essential for quality control, sorting, and reducing post-harvest losses in the agricultural and food supply chain. This study proposes a deep learning-based image classification approach to categorize bananas into four ripeness stages: overripe, ripe, rotten, and unripe. Two convolutional neural network (CNN) architectures were developed and compared: a CNN built from scratch and a MobileNetV2 model utilizing transfer learning with fine-tuning. The dataset, consisting of thousands of labeled banana images across the four classes, was split into training, validation, and testing sets. Experimental results show that the CNN from scratch achieved a training accuracy of 97.52%, validation accuracy of 97.48%, and testing accuracy of 97.26%, while the MobileNetV2 model achieved higher performance with a training accuracy of 99.65%, validation accuracy of 99.18%, and testing accuracy of 98.96%. Both models were categorized as good fit based on their training-validation accuracy gap. The MobileNetV2 transfer learning model outperformed the CNN from scratch and was selected as the best-performing model. The final model was exported into three deployment-ready formats—SavedModel, TensorFlow Lite, and TensorFlow.js—to support real-time and cross-platform implementation. These findings demonstrate that transfer learning with MobileNetV2 provides a more accurate and efficient solution for automated banana ripeness classification compared to a custom CNN architecture.

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1. INTRODUCTION

Bananas are among the most widely consumed and economically important fruit commodities in tropical and subtropical regions, where the fruit's ripeness stage directly determines its market value, taste quality, and shelf life. Because post-harvest ripening is a continuous process, produce distributors, retailers, and food processing industries require reliable methods to classify ripeness stages so that sorting, storage, and distribution decisions can be made efficiently and losses can be minimized since ripening of banana hands during handling, transit, shipping, and storage leads to higher post-harvest loss and impedes trade, making ripeness identification a matter of paramount importance for bulk handlers and food processing industries. Traditionally, ripeness assessment has relied on manual visual inspection performed by trained workers. This approach is inherently subjective, time-consuming, and difficult to scale, motivating the development of automated, non-destructive classification methods based on computer vision.

In recent years, deep learning—particularly Convolutional Neural Networks (CNN)—has become the dominant approach for fruit ripeness classification because of its ability to automatically learn discriminative visual features such as color, texture, and surface patterns directly from images. Machine learning and deep learning approaches to banana ripeness classification represent a significant advancement in agriculture and the agri-food industry by enabling optimized decision-making from harvest to distribution, and recent studies applying CNN-based architectures have achieved notably high classification accuracy. Several prior works illustrate the range of approaches explored in this domain. Saragih and Emanuel developed a four-layer CNN model to classify bananas into seven ripening stages using the Mazen and Nashat dataset, incorporating a parameter optimization approach[1]. Other research proposed a hybrid CNN combined with an eXtreme Gradient Boosting algorithm to classify banana fruit into four ripeness categories—unripe, underripe, ripe, and overripe—using a custom-built dataset. More recent studies have compared multiple pretrained architectures directly; for example, one study found that a DenseNet-121 model achieved 93.02% accuracy in classifying banana ripening stages, outperforming a ResNet-50 model at 92.44%, while another compared a proposed nine-layer CNN against VGG16, VGG19, InceptionV3, ResNet50, and EfficientNetB0 on a dataset of Nendran bananas, reporting that the custom CNN achieved 95% accuracy and outperformed the pretrained alternatives[2]. Transfer learning approaches have also been explored to reduce data and computation requirements; one study combined synthetic and real banana ripeness data with a simple CNN architecture improved through transfer learning to classify real-world images, although findings on fine-tuning sensitivity have shown that adjusting the unfreezing layer can noticeably affect

performance, with MobileNetV2 in particular showing accuracy drops from 94.66% to 91.60% under certain configurations.

Despite this growing body of work, most existing studies either evaluate a single CNN architecture in isolation or compare several pretrained networks without directly contrasting a custom-built CNN against a transfer-learning-based model trained under equivalent experimental conditions. Furthermore, few studies extend their evaluation beyond offline accuracy metrics to consider deployment readiness, even though real-world adoption of ripeness classification systems—whether in mobile sorting applications, embedded devices, or web-based inspection tools—depends on models that can be exported into lightweight, platform-compatible formats[3].

To address this gap, this study presents a comparative analysis between a Convolutional Neural Network built from scratch and a MobileNetV2 model utilizing transfer learning with fine-tuning for the task of classifying banana ripeness into four categories: overripe, ripe, rotten, and unripe. Both models are trained and evaluated on the same dataset split to ensure a fair comparison across training, validation, and testing accuracy, as well as confusion matrix and per-class classification performance[4]. The best-performing model is then exported into three deployment-ready formats—SavedModel, TensorFlow Lite, and TensorFlow.js—to support practical implementation across server, mobile, and web-based environments. The contributions of this study are twofold: (1) a systematic performance comparison between a custom CNN architecture and a transfer-learning-based MobileNetV2 model under identical experimental conditions, and (2) a deployment-oriented pipeline that automatically selects and exports the best-performing model across multiple inference-ready formats[5].

Beyond raw classification accuracy, several studies emphasize that the practical value of a banana ripeness classification system also depends on how ripeness assessment methods are categorized and deployed in real-world conditions. A systematic review of banana ripeness detection methods distinguishes between destructive approaches, which require physically cutting or peeling the fruit and are therefore costly and dependent on expert skill, and non-destructive approaches, which rely on external physical and spectral properties such as color and texture and are consequently far more practical for large-scale, automated sorting systems[6]. Within the non-destructive category, CNN-based methods have become the preferred choice due to their ability to learn ripeness-related visual cues without manual feature engineering, yet the computational cost of deep architectures remains a limiting factor for deployment on mobile devices, embedded systems, or web-based platforms commonly used in small- and medium-scale agricultural operations. This consideration motivates the use of lightweight, pretrained architectures such as MobileNetV2, which is specifically designed to balance classification accuracy with computational efficiency through depthwise separable convolutions, making it well suited for transfer learning in resource-constrained deployment scenarios. Building on these considerations, the remainder of this paper is organized as follows: Section 2 describes the dataset and the proposed methodology, including the CNN-from-scratch and MobileNetV2 transfer-learning architectures; Section 3 presents the experimental results and comparative analysis; and Section 4 concludes the paper and outlines directions for future work.

2. RESEARCH METHOD

2.1 Dataset Distribution

The dataset used in this study is the publicly available *Banana Classification* dataset (sourced via KaggleHub, atrithakar/banana-classification), consisting of images of bananas labeled across four ripeness categories: overripe, ripe, rotten, and unripe. The original dataset folders were merged and re-split into training, validation, and testing sets using an 80:10:10 ratio with stratified sampling to preserve class proportions across all subsets. The resulting distribution is summarized in Table 1.

Class	Training	Validation	Testing	Total	Proportion
Overripe	2,153	269	269	2,691	19.96%
Ripe	3,212	401	402	4,015	29.78%
Rotten	3,674	460	459	4,593	34.07%
Unripe	1,743	218	218	2,179	16.17%
Total	10,783	1,349	1,349	13,481	100%

Table 1. Dataset distribution across training, validation, and testing sets

As shown in Table 1, the dataset exhibits a moderate class imbalance, with the *rotten* class being the most represented (34.07%) and the *unripe* class being the least represented (16.17%). This distribution was taken into account during model evaluation, particularly in the interpretation of per-class precision and recall in Section III.

2.2 Data Preparation

Prior to training, all images were preprocessed through the following pipeline:

1. **Resizing** — all images were resized to a fixed input dimension of 224×224 pixels to match the input requirements of both the CNN-from-scratch and the MobileNetV2 architecture.
2. **Normalization** — pixel values were rescaled to the [0, 1] range for the CNN-from-scratch model, while the MobileNetV2 model used the architecture-specific preprocess_input normalization to match its ImageNet pretraining statistics.
3. **Data augmentation** — to improve generalization and reduce overfitting given the moderate dataset size, on-the-fly augmentation was applied to the training set, including random rotation, horizontal flipping, and slight perspective/shear transformation. This is consistent with the varied orientations observed in the inference samples (Figure 9), where banana images appear at different rotation angles and framing.
4. **Batching and shuffling** — the training set was shuffled at each epoch and grouped into mini-batches to stabilize gradient updates.

2.3 CNN-from-Scratch Architecture

The first model is a custom Convolutional Neural Network trained from scratch, without any pretrained weights. The architecture consists of four convolutional blocks with increasing filter depth, followed by a global pooling layer and fully connected classification layers, as illustrated in Figure 10.

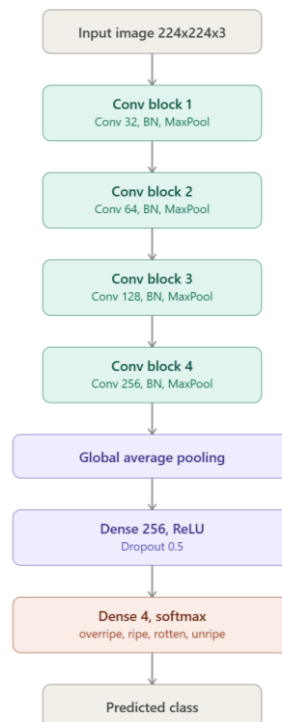


Figure 1. CNN-from-scratch architecture

Training was conducted for 28 epochs using categorical cross-entropy loss and the Adam optimizer, with early stopping based on validation loss to prevent overfitting.

2.4 Transfer Learning Architecture (MobileNetV2)

The second model leverages MobileNetV2 pretrained on ImageNet as a feature extractor. Training was performed in two stages: (1) a feature-extraction stage in which the MobileNetV2 base was frozen and only the newly added classification head was trained, and (2) a fine-tuning stage in which the top layers of the base network were unfrozen and retrained jointly with the classifier head at a lower learning rate to adapt the pretrained features to the banana ripeness domain. The overall architecture is shown in Figure 11.

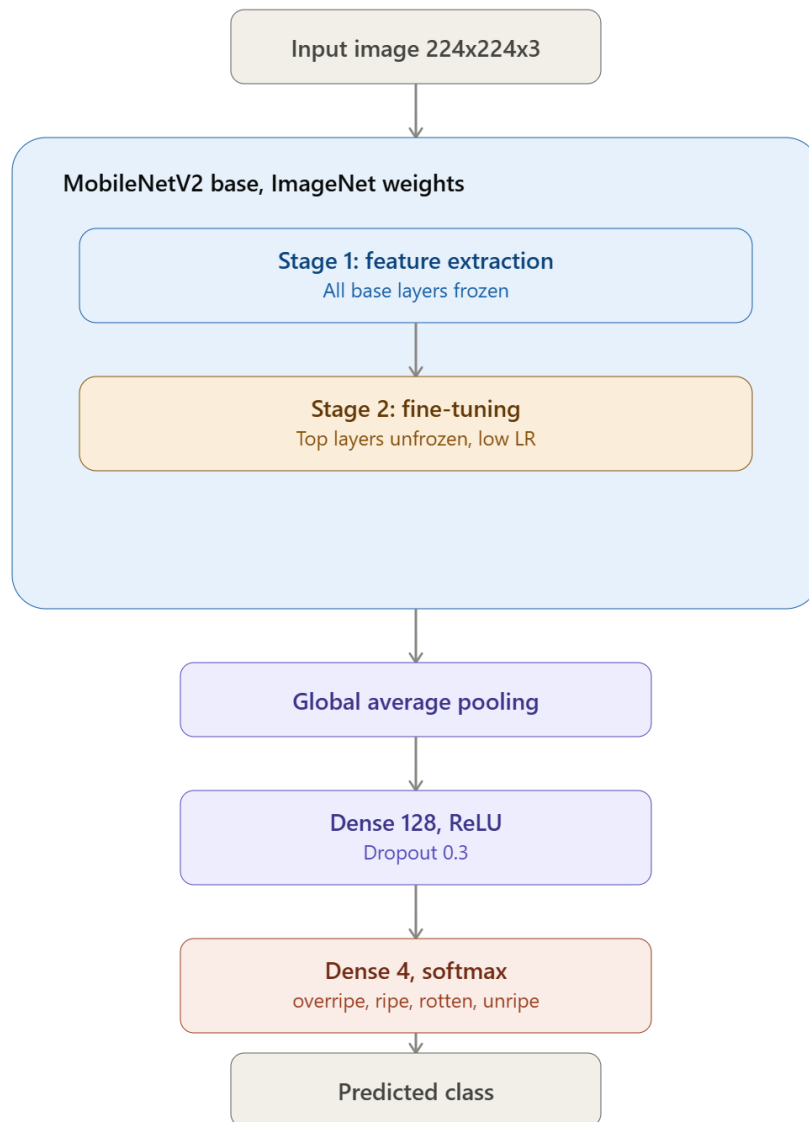


Figure 2. MobileNetV2 transfer learning architecture, trained in two stages

Training was carried out over 54 epochs in total across both stages, using categorical cross-entropy loss and the Adam optimizer with a reduced learning rate during the fine-tuning stage to avoid catastrophic forgetting of the pretrained ImageNet features.

2.5 Evaluation Metrics

Accuracy:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision:

$$Precision = \frac{TP}{TP + FP}$$

Recall:

$$Recall = \frac{TP}{TP + FN}$$

F1-score:

$$F1 - score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

In addition to these per-class metrics, each model was categorized as *good fit*, *overfit*, or *underfit* based on the gap between training and validation accuracy across epochs, following the criterion that a small, stable gap (as observed in both models in this study; see Section III) indicates a good fit, whereas a large and widening gap would indicate overfitting.

3. RESULTS AND DISCUSSION

3.1 Training Results

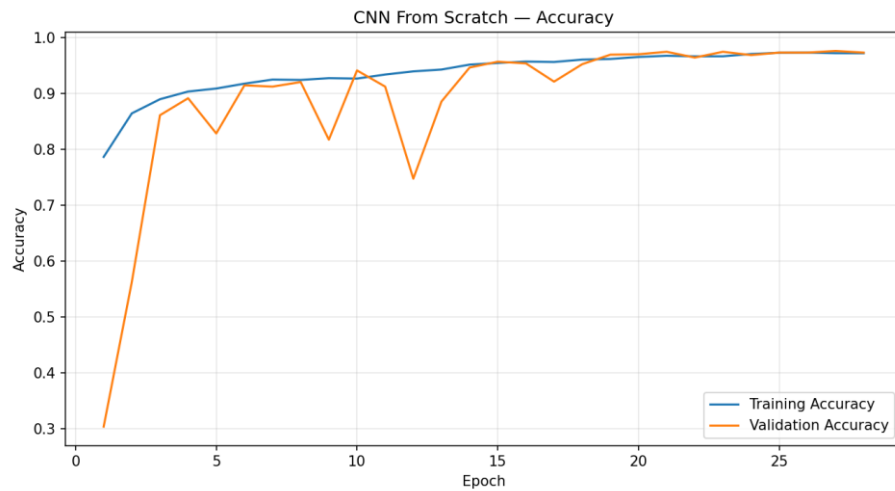


Figure 3. Training and validation accuracy of the CNN-from-scratch model

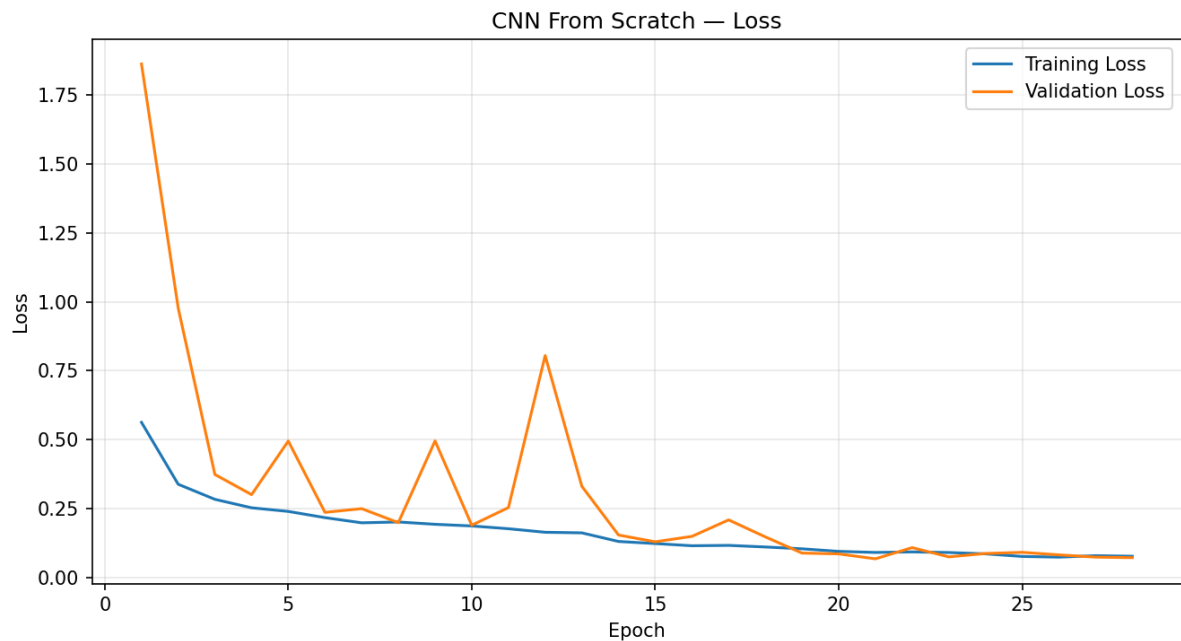


Figure 4. Training and validation loss of the CNN-from-scratch model

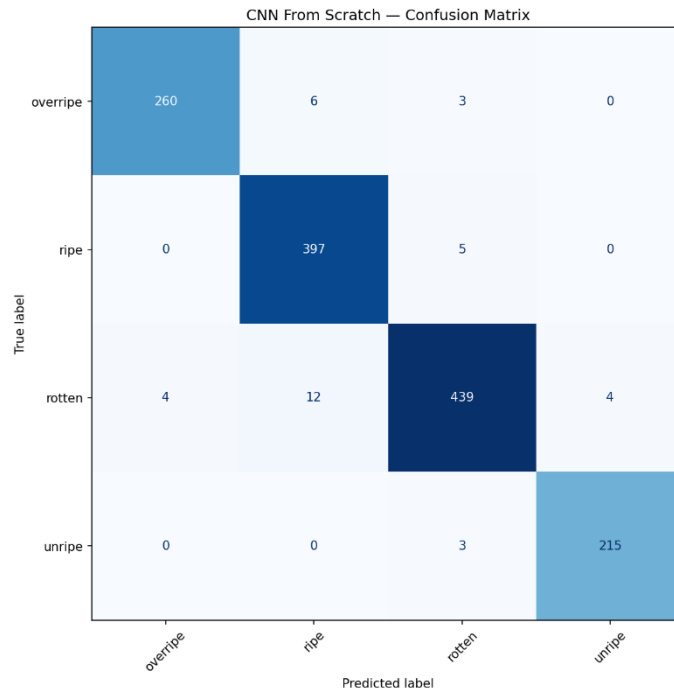


Figure 5. Confusion matrix of the CNN-from-scratch model on the testing set

present the training and validation curves for the CNN-from-scratch model over 28 epochs. The validation accuracy exhibited considerable fluctuation during the early epochs, rising sharply from around 30% at epoch 1 to roughly 86% by epoch 3, and dipping notably at epochs 5, 9, and 12 (with validation accuracy briefly falling to approximately 75% at epoch 12, mirrored by a corresponding validation loss spike to about 0.80). These fluctuations are consistent with a model still learning discriminative features from scratch without pretrained priors. From epoch 15 onward, both training and validation curves stabilize and converge closely, with the training loss steadily decreasing to approximately 0.07 and validation loss to approximately 0.08 by the final epoch, indicating that the model reached a stable convergence point without diverging or overfitting.

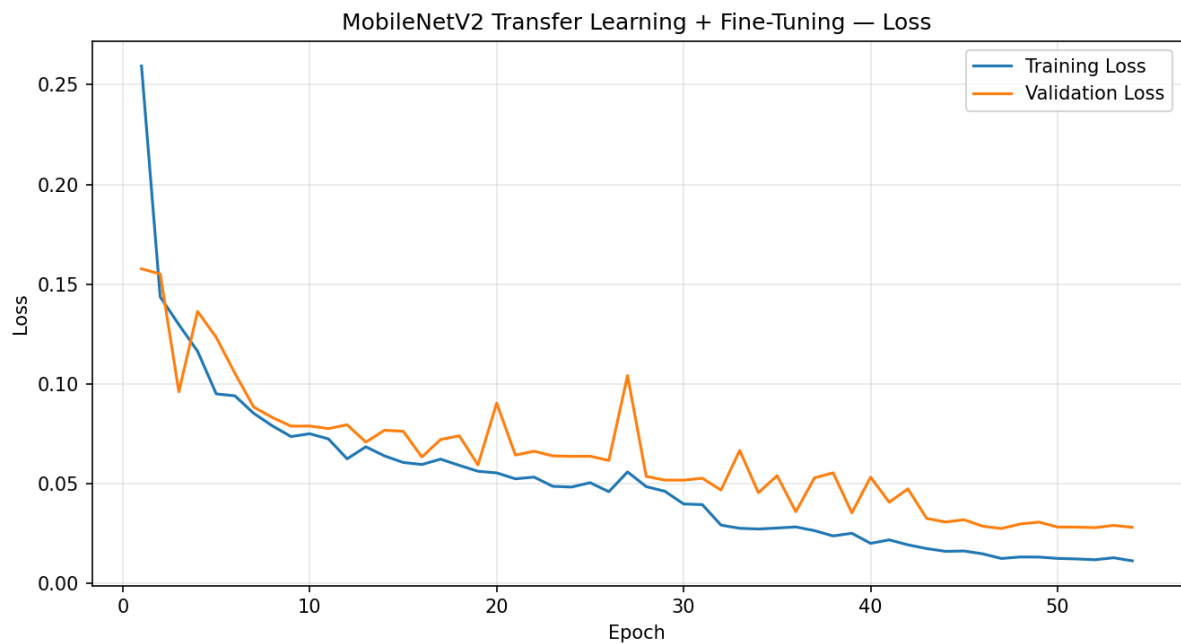


Figure 6. Training and validation loss of the MobileNetV2 transfer learning + fine-tuning model

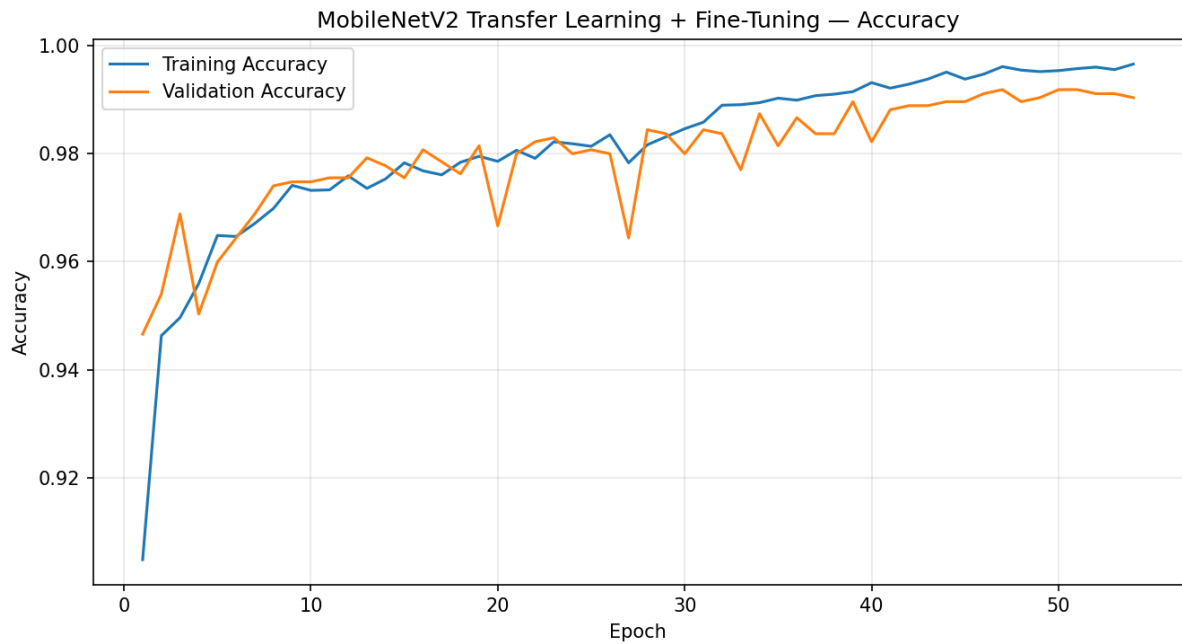


Figure 7. Training and validation accuracy of the MobileNetV2 transfer learning + fine-tuning model

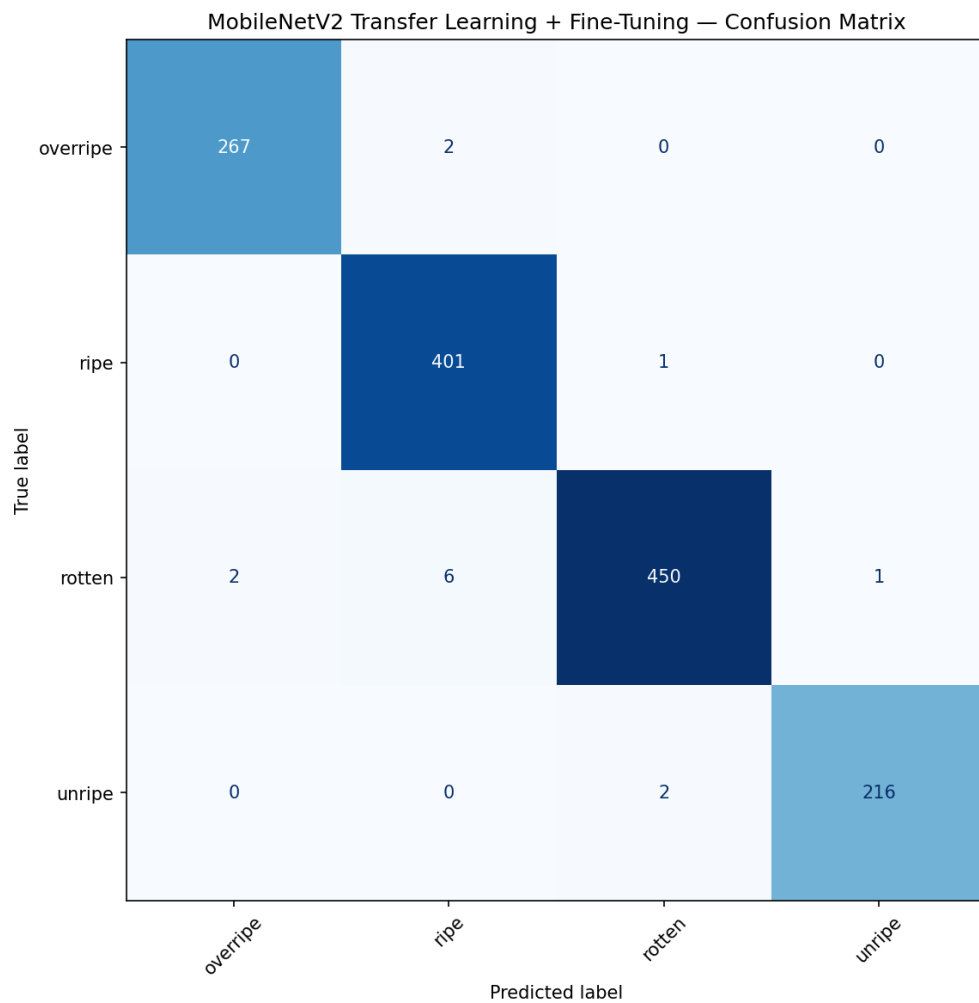


Figure 8. Confusion matrix of the MobileNetV2 transfer learning + fine-tuning model on the testing set

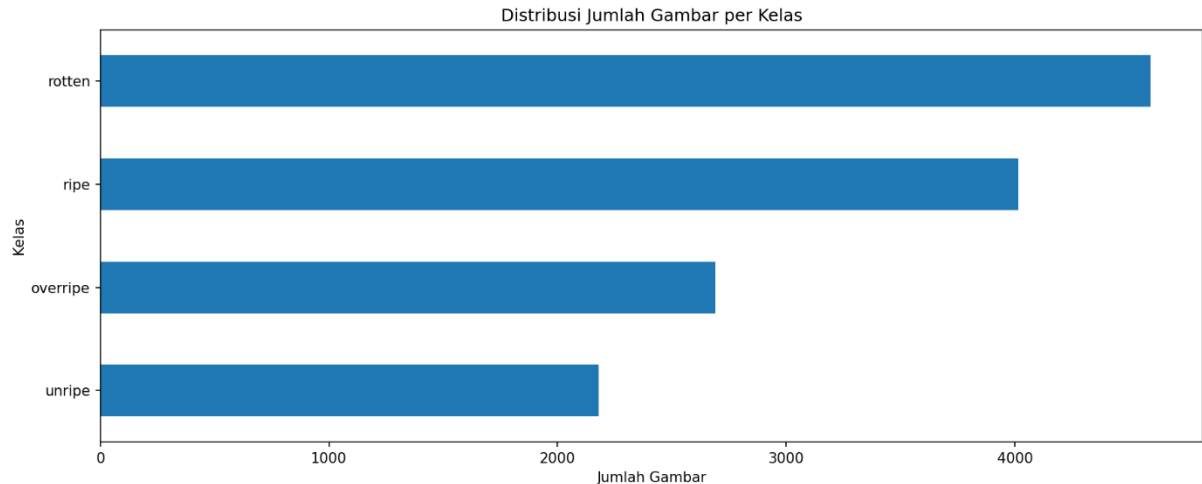


Figure 9. Class distribution of the banana ripeness dataset

show the training and validation curves for the MobileNetV2 transfer learning model over 54 epochs (covering both the feature-extraction and fine-tuning stages). In contrast to the CNN-from-scratch model, MobileNetV2 started from an already high validation accuracy of approximately 94.7% at epoch 1, reflecting the benefit of ImageNet-pretrained features. The curve rises smoothly with only minor fluctuations (small dips around epochs 20 and 27) before converging to a training accuracy of 99.65% and validation accuracy of 99.18%. The training and validation loss curves remain closely aligned throughout, decreasing from approximately 0.26/0.16 to 0.01/0.03 respectively, indicating strong generalization with minimal overfitting despite the very high accuracy achieved.

3.2 Testing Results

Table 2 presents the per-class classification report for the CNN-from-scratch model on the testing set, while **Table 3** presents the corresponding report for the MobileNetV2 model.

Class	Precision	Recall	F1-score	Support
Overripe	0.985	0.967	0.976	269
Ripe	0.957	0.988	0.972	402
Rotten	0.976	0.956	0.966	459
Unripe	0.982	0.986	0.984	218
Accuracy			0.973	1,348
Macro avg	0.975	0.974	0.974	1,348
Weighted avg	0.973	0.973	0.973	1,348

Table 2. CNN-from-scratch classification report (testing set)

Class	Precision	Recall	F1-score	Support
Overripe	0.993	0.993	0.993	269
Ripe	0.980	0.998	0.989	402
Rotten	0.993	0.980	0.987	459
Unripe	0.995	0.991	0.993	218
Accuracy			0.990	1,348
Macro avg	0.990	0.990	0.990	1,348
Weighted avg	0.990	0.990	0.990	1,348

Table 3. MobileNetV2 transfer learning classification report (testing set)

Based on the confusion matrix in **Figure 5**, the CNN-from-scratch model most frequently confused the *rotten* class with *ripe* (12 out of 459 rotten samples misclassified), followed by minor confusion between *overripe* and *ripe/rotten*. This is consistent with the visual similarity between late-stage ripening and early spoilage, where surface browning patterns overlap between the *rotten* and *ripe/overripe* categories. The *unripe* class was the most reliably distinguished, with only 3 misclassifications out of 218 samples, owing to its visually distinct green coloration.

The MobileNetV2 confusion matrix in **Figure 8** shows a similar but substantially reduced confusion pattern: the *rotten* class remains the most challenging (6 misclassified as *ripe*, 2 as *overripe*, 1 as *unripe*, out of 459 samples), while the *overripe*, *ripe*, and *unripe* classes achieved near-perfect classification (267/269, 401/402, and 216/218 respectively). This indicates that the pretrained MobileNetV2 features generalize more effectively to the subtle visual boundary between ripening stages than the features learned by the CNN from scratch.

3.3 Comparative Analysis

Table 4 summarizes the overall performance comparison between the two models across training, validation, and testing sets.

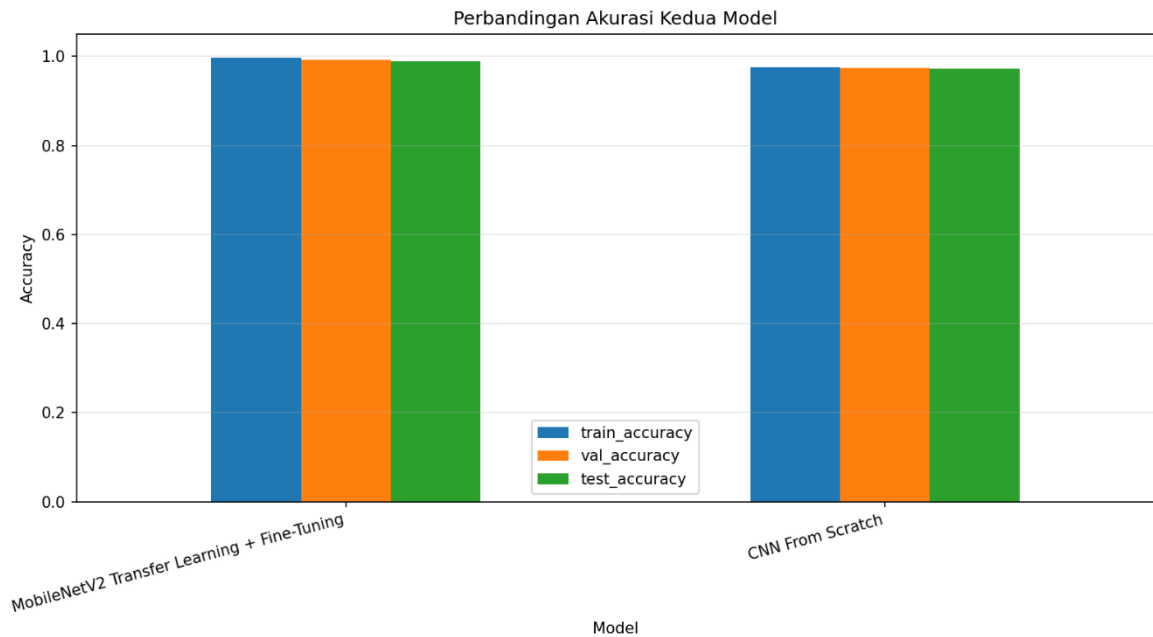


Figure 10. Comparison of training, validation, and testing accuracy between the two models

	Train Loss	Train Acc.	Val Loss	Val Acc.	Test Loss	Test Acc.	Fit Status
CNN From Scratch	0.0725	97.52%	0.0686	97.48%	0.0813	97.26%	Good Fit
MobileNetV2 (TL + FT)	0.0126	99.65%	0.0276	99.18%	0.0362	98.96%	Good Fit

Table 4. Performance comparison between CNN-from-scratch and MobileNetV2 transfer learning

Both models were categorized as *good fit*, as the gap between their training and validation accuracy remained small and stable throughout training (0.04–0.47 percentage points), indicating neither model suffered from significant overfitting or underfitting. However, MobileNetV2 consistently outperformed the CNN from scratch across all three data splits, achieving a 1.70 percentage-point higher testing accuracy (98.96% vs. 97.26%) and a lower testing loss (0.0362 vs. 0.0813). This result confirms that transfer learning from ImageNet-pretrained weights, combined with fine-tuning, provides a measurable advantage over training a comparable CNN architecture from scratch on this dataset size (13,481 images), likely because the pretrained convolutional filters already encode general texture and color features that transfer well to ripeness-related visual cues. Based on testing accuracy, **MobileNetV2 Transfer Learning + Fine-Tuning was selected as the best-performing model** and used for deployment export.

3.4 Deployment and Inference Evidence

Following model selection, the best-performing model (MobileNetV2) was exported into three deployment-ready formats: TensorFlow SavedModel (for server-side inference), TensorFlow Lite (.tflite, for mobile and embedded deployment), and TensorFlow.js (for browser-based inference), with the export pipeline automatically selecting and naming the winning model based on testing accuracy.



Figure 11. Sample inference results on test images using the exported model

To validate the exported TensorFlow Lite model, inference was performed on nine sample images spanning all four ripeness classes, shown qualitatively in **Figure 11**. However, this deployment test revealed a discrepancy between the original Keras model's performance and the exported TFLite model's behavior: as summarized in **Table 5**, the TFLite model predicted every one of the nine samples as *overripe*, regardless of the true label, and returned an undefined (NaN) confidence score for each prediction.

True label	Predicted label	Confidence	Correct
Rotten	Overripe	NaN	X
Overripe	Overripe	NaN	✓
Rotten	Overripe	NaN	X
Unripe	Overripe	NaN	X
Ripe	Overripe	NaN	X
Unripe	Overripe	NaN	X
Ripe	Overripe	NaN	X
Overripe	Overripe	NaN	✓
Ripe	Overripe	NaN	X

Table 5. TFLite inference validation on nine sample images

Only the two genuinely *overripe* samples were classified correctly (2/9, 22.2%), a result far below the 98.96% testing accuracy achieved by the source Keras model, and the constant NaN confidence score confirms a systematic

pipeline error rather than a model quality issue. This pattern — a single dominant output class combined with undefined probability values — is a known symptom of a preprocessing or quantization mismatch between the training pipeline and the TFLite inference pipeline, such as an input tensor not being normalized/scaled consistently with `preprocess_input`, an incorrect output tensor dequantization step, or a data type mismatch between the model's expected input and the image array supplied at inference time. This finding highlights an important gap between offline model accuracy and real-world deployment readiness, and indicates that the TFLite conversion and inference script require further debugging — specifically verifying the input preprocessing and output tensor handling — before the exported model can be considered production-ready.

CONCLUSION

This study presented a comparative analysis between a Convolutional Neural Network built from scratch and a MobileNetV2 model utilizing transfer learning with fine-tuning for classifying banana ripeness into four categories: overripe, ripe, rotten, and unripe. Both models achieved strong performance and were categorized as good fit, with training-validation accuracy gaps remaining small and stable throughout training. However, the MobileNetV2 transfer learning model consistently outperformed the CNN from scratch across all data splits, achieving a testing accuracy of 98.96% compared to 97.26% for the CNN from scratch, along with a lower testing loss and fewer misclassifications in the confusion matrix — particularly for the *rotten* class, which was the most challenging category for both models due to its visual similarity with *ripe* and *overripe* stages. These results directly address the objective of this study by demonstrating that leveraging ImageNet-pretrained features through transfer learning provides a measurable and consistent advantage over training a comparable architecture from scratch, even on a moderately sized dataset of 13,481 images, and confirm MobileNetV2 Transfer Learning + Fine-Tuning as the best-performing model for this classification task.

Beyond offline accuracy, this study also evaluated deployment readiness by exporting the best-performing model into three formats — SavedModel, TensorFlow Lite, and TensorFlow.js. This deployment evaluation revealed an important limitation: the exported TensorFlow Lite model produced degenerate predictions (a single dominant output class with undefined confidence scores) during inference testing, indicating a preprocessing or quantization mismatch between the training and inference pipelines rather than a deficiency in the underlying model. This finding underscores that high offline accuracy alone does not guarantee reliable real-world deployment, and that export pipelines require independent validation before production use.

Based on these findings, several directions are recommended for future work. First, the TensorFlow Lite conversion and inference pipeline should be debugged and revalidated — particularly the input normalization and output tensor dequantization steps — to ensure consistency with the original Keras model before deployment. Second, future studies could expand the dataset to include greater variation in lighting conditions, backgrounds, and banana varieties to improve model robustness under real-world conditions. Third, given the persistent confusion between the *rotten* and *ripe/overripe* classes, future work could explore additional input modalities (e.g., color histogram features or multi-angle imaging) or explainability techniques such as Grad-CAM to better understand and reduce this confusion. Finally, further comparison with other lightweight architectures (e.g., EfficientNet-Lite, MobileNetV3) is recommended to identify the optimal trade-off between accuracy and inference efficiency for real-time deployment on mobile and embedded devices.

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